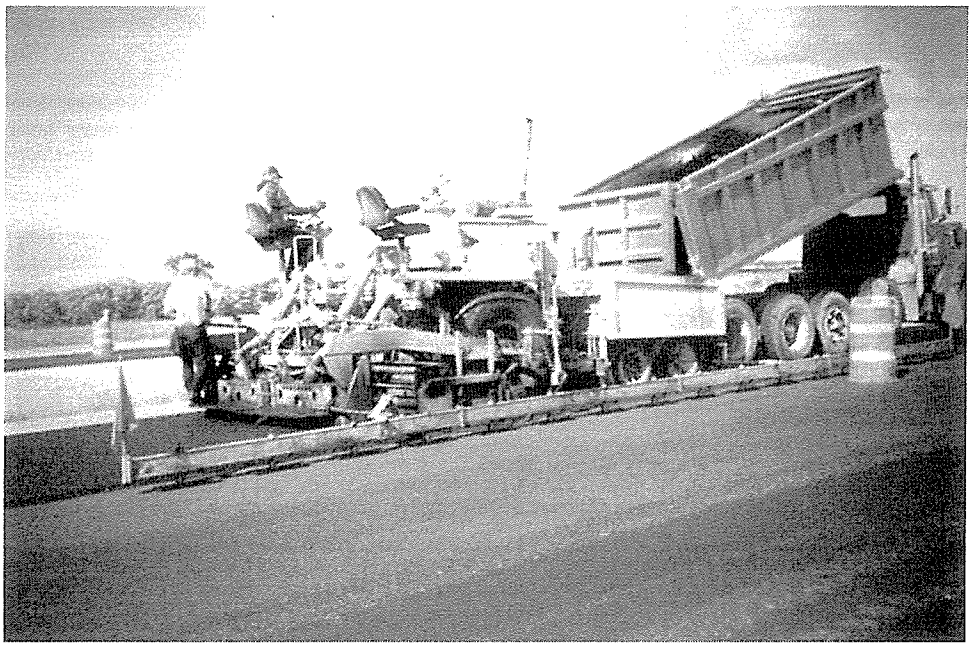


Figure 8.21 Thirty-Foot Mobile Reference

beam. The feet are spring-loaded so that they can be moved by a small undulation or rock on the roadway without altering the slope of the beam. The floating-beam reference system will average the variation of the existing grade over 30 or 40 feet (9 or 12 meters).

- **Sonic trackers.** The sonic-tracker mobile reference system mounts a rigid beam on the side of the paver with four sensors mounted on it (see Figure 8.22). The sensors direct sonic waves downward and times the return echo to establish the height of the sensor above the reference surface. The four height readings are continuously averaged together to provide a reference line.
- **Curb and gutter.** When HMA pavement is being placed against a curb and/or gutter, either the curb or the gutter can be used as a reference for controlling the grade side of the screed. If the sensor is mounted just ahead of the augers indexing off the curb or gutter, it will accurately hold the depth of the HMA in relation to the curb or gutter.

8.4.5 Leveling Effect of the Paver and the Free-Floating Screed

The Line of Pull—Wheelbase of the Paver—Length of the Side Arm

Either style of paver (rubber-tire or track-mounted) utilizes the wheelbase of the paver as the first step in achieving a smooth HMA pavement over a

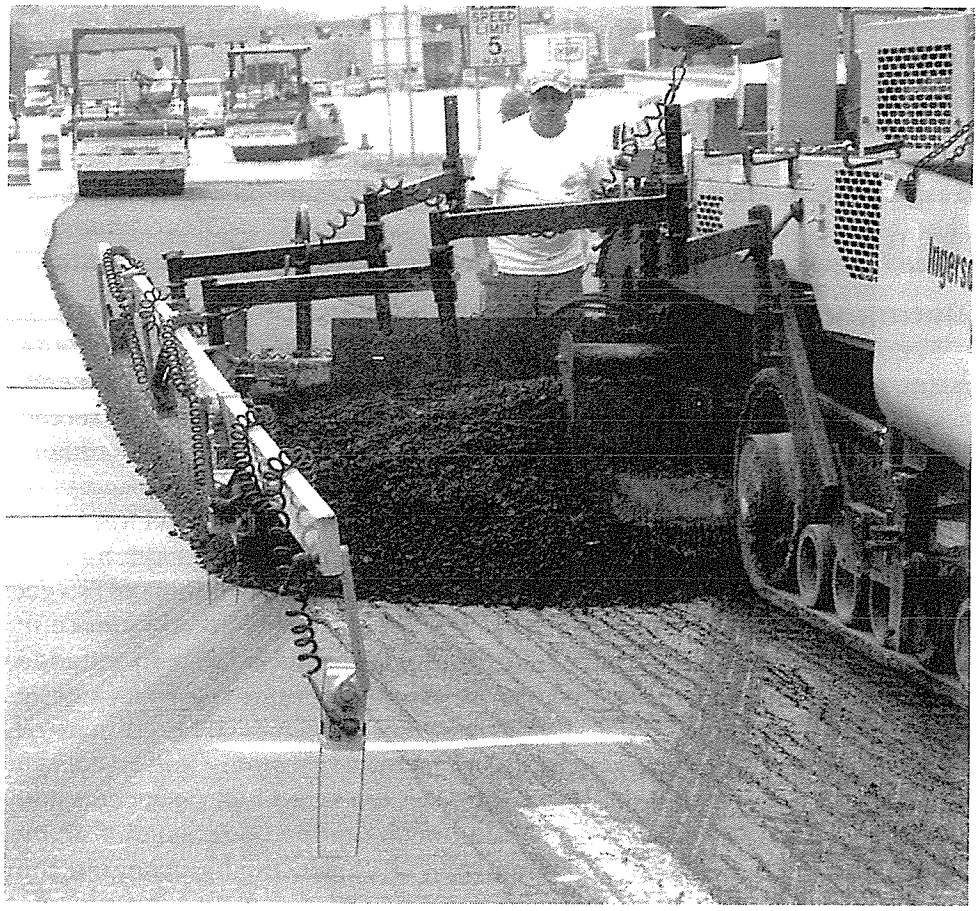


Figure 8.22 Sonic Sensor Mobile Reference System

rough base. The wheelbase of the paver develops what is called the "line of pull," which can be thought of as the vertical center of gravity of the paver traced longitudinally along the path of the paver. Vertically lifting only the front or the back of a paver moves the center of gravity of the paver up approximately one-half the distance lifted. When the tow point of the screed is located near this central fulcrum point, this averaging effect makes proportionally smaller changes in the height of the tow point and explains how the paver's wheel base is used to level uneven surfaces. The small front wheels or bogies under the hopper that can move across a depression or a ridge will only reflect a small amount of vertical movement at the tow point. When the drive tires in the rear pass over this same point, the vertical movement is also reduced. Track pavers also provide this leveling effect. They have a series of small bogie wheels that are free to move up and down independent of one another as they run inside of the flexible continuous track.

The side arms, attached on each side of the paver, vary slightly among the paver manufacturers, but overall they are about 10 feet (3 meters) in length. The side arms attach to the paver at the tow point, which defines

the line of pull (see above) as the paver moves forward. Using the line of pull as a reference line, any change in the vertical height of the line of pull will be reduced by an 8:1 ratio when it reaches the screed and changes the angle of attack. For example, if the line of pull changes in height by 1 inch (25 millimeters), the screed angle of attack, will change only ⅛ inch (3 millimeters). The leveling effect of the paver's running gear, combined with the 8:1 reaction ratio between the line of pull and the change in screed height, explains why HMA pavements can be extremely smooth.

Distance for Screed Change to Occur

When adjusted properly, the forces acting on a free-floating screed are balanced and it will lay a smooth, high quality HMA pavement over a rough base. As previously discussed, when the forces acting on the screed become unbalanced, the depth of the HMA pavement changes gradually as the paver moves forward and the screed again seeks equilibrium with the forces acting upon it. The distance the paver must travel to reach this equilibrium is generally 4 to 5 times the length of the side arms of the paver (see Figure 8.23). With the typical side arm length of 10 feet (3 meters), the paver will move 40 to 50 feet (12 to 15 meters) before the screed is again in equilibrium. This gradual transition over an extended length should eliminate the need for the screed operator to crank in large adjustments on the manual depth-screw mechanisms. When a paver is properly set up and paving continuously without stopping, the screed will remain in balance and the depth screws on the screed will not need manual adjustments. When paving with automatic grade controls and when a depth change is needed, the change can be "dialed in" by changing the depth setting in the grade control box.

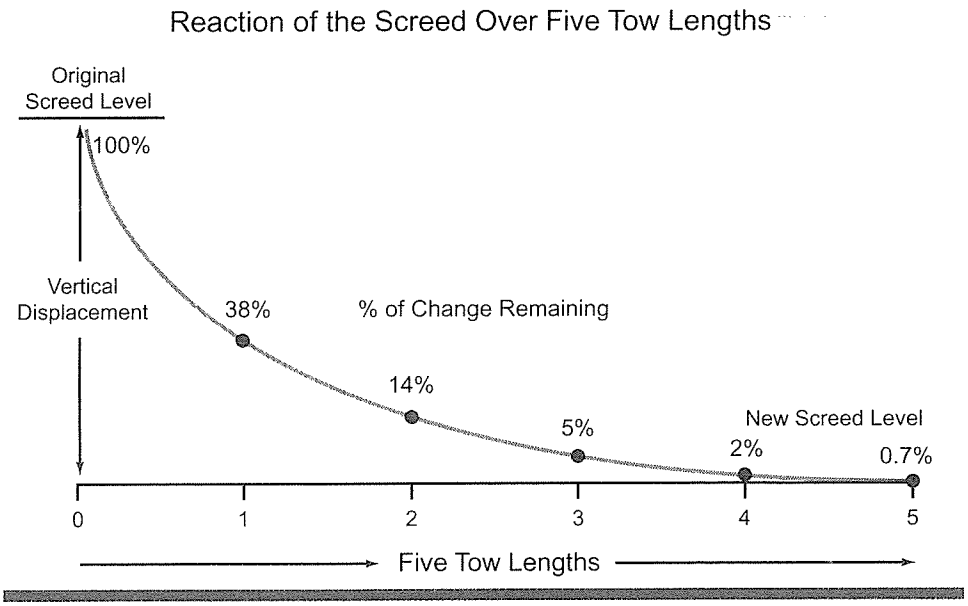


Figure 8.23 Distance Required for Screed Change to Fully Occur

Establishing the Proper Paving Speed

Planning is necessary for achieving a high quality asphalt pavement. Planning begins with establishing the amount of material to be produced, the number of trucks needed, the hauling distance, and the amount of HMA that can be placed and compacted before the temperature falls below compaction levels. The most common paver operator error is excessive paver speed. The paver operator must monitor the arrival rate of the material to the job site, the flow of material through the paver, and the pace of the compaction train. Paver speed should be closely monitored to match the consumption rate of material through the paver with the HMA production and delivery.

Calculating Paver Yield

During HMA placement, the paving supervisor and/or the inspector should periodically check the yield of the paving operation. The purpose is to cross-check the thickness measurements to ensure that the HMA is being placed at the proper average thickness. Some agencies use paver yield (or spread rate) as an acceptance criteria in lieu of thickness. This check can be made on each truckload of HMA, but adjustments to the thickness should not be made that often. A longer distance (1,000 feet; 300 meters) may be more appropriate.

The first step in calculating paver yield is to determine the post-compaction unit weight (pounds per square yard; kilograms per square meter) of the HMA. This can be determined from the bulk-specific gravity from previous cores or from nuclear density measurements. The post-compaction unit weight of the HMA and the mat's thickness, width, and length can be used to calculate the yield. The following is an example problem and its solution:

Problem

A series of trucks has delivered 425.5 tons (386.0 tonnes) of HMA to the project. This HMA covered 3,500 feet (1,066.9 meters) of pavement. The paver is placing the HMA 12 feet (3.66 meters) wide and 1½ inches (38.1 millimeters) thick (compacted). The HMA has a unit weight of 151.5 pounds per cubic foot (2,426.8 kilograms per cubic meter).

Solution (English units)

Calculate the *required* HMA weight:

$$\frac{(\text{length} \times \text{width} \times \text{depth} \times \text{in-place density})}{(2{,}000 \text{ pounds/ton})}$$

$$\frac{(3{,}500 \text{ feet}) \times (12 \text{ feet}) \times (0.125 \text{ foot}) \times (151.5 \text{ pounds per cubic foot})}{(2000 \text{ pounds/ton})}$$

$$= 397.7 \text{ tons of HMA}$$

Therefore, 7.0 percent more HMA than needed is being placed (425.5/397.7) or an average of 1.60 inches (versus the 1.5 inches required).

Solution (Metric units)
Calculate the required HMA weight:

$$\frac{(\text{length} \times \text{width} \times \text{depth} \times \text{in-place density})}{(1000 \text{ kg/metric tonne})}$$

$$\frac{(1{,}066.9 \text{ meters}) \times (3.66 \text{ meters}) \times (0.381 \text{ meters}) \times (2{,}426.8 \text{ kg/m}^3)}{(1000 \text{ kg/metric tonne})}$$

$$= 361.1 \text{ metric tonnes of HMA}$$

Therefore, approximately 6 percent more HMA than needed is being placed (386.0/361.1) or an average of 40.7 millimeters (versus the 38.1 millimeters required.)

8.4.6 Fumes Control

Fumes control systems are designed to draw the excess heat and fumes generated in the auger area away from the workers and exhaust it above their heads or draw it into the intake manifold of the paver's engine. Baffles, ductwork, and fans are used to capture the excess heat and fumes and move them up and out of the work area. Eighty-five percent of fumes generated in the auger area are captured and exhausted out of the work area. Maintaining this level of control demands that the fumes control system be properly maintained and operated.

8.4.7 Observation of the HMA Pavement Behind the Screed

The asphalt in the head of material in the auger area must flow without hesitation under the screed into the HMA pavement being laid. Any cooler or slightly compacted chunk of asphalt that finds its way into this area must be broken up before it passes under the screed. This breaking-up process disrupts the normal flow of HMA under the screed and results in a dragging or tearing of the HMA pavement surface. Identifying and eliminating the source of the chunks is the only solution to eliminating these HMA pavement blemishes. Segregation is another HMA pavement fault revealed by observation. Longitudinal streaks of segregation usually indicate an auger-feeding problem. An accumulation of segregated material in and around the center auger drive box can result in a streak of segregated material 6 to 8 inches (150 to 200 millimeters) wide appearing in the longitudinal center mat. Readjusting the volume of HMA in this area or slightly raising the augers can improve or minimize these segregated streaks. Excessive material pulled under the paver can cause similar segregated streaks on the surface or on the bottom of the mat. Properly setting the material sensors and reducing the volume of the HMA in the auger area can reduce the amount of material pulled under the belly of the paver.

Roughness

Roughness is difficult to troubleshoot. The type of roughness must be determined in order to find a solution. Examples of roughness include short or long swells, ripples, load-to-load swells, or random roughness.

Short swells or ridges, sometimes referred to as ripples, are usually the result of the screed's angle of attack changing on its own during the paving operation. If the screed is worn or otherwise shows signs of being loose at its connections to the side arms, it can allow the angle of attack to change uncontrollably and cause these ripples. The screed-depth screws should be checked for looseness—there should be no more than one-half turn of free rotary motion when the screed is resting on a flat surface and both depth screws are in the null position (zero angle of attack). The bearings, fasteners, and screed pivot pins should also be checked for looseness. Corrections should be made to eliminate angle-of-attack changes that are not introduced by the automatic grade and slope or by the manual adjustment of the depth screws.

Swells that are longer than the length of the paver can usually be attributed to a change in the volume of the head of material. If the head of material in front of the screed is allowed to vary by more than ±1 inch (25 millimeters), this type of roughness can appear. Truck exchanges should be watched carefully. The time it takes one truck to exit the paver hopper and the next one to get into the discharge position can be long enough that the flow of material going to the auger area can be interrupted. This happens because the head of material is not being replenished, allowing the screed to drop. If this happens, roughness will result with each truck exchange. This type of roughness is also known as long swells. The solution is to train the truck driver and the paver operator how to exchange trucks without varying the head of material. If everyone works together and communication between paver operator, the dump operator (if used), and the truck driver is good, it can be done. The screed operator should monitor the head of material and let everyone know if the volume varies more than ±1 inch (25 millimeters). Long swells can also occur if the stringline being used as a grade reference is allowed to sag between supports or the grade sensors get out of adjustment. Constantly checking the tension in the stringline and monitoring the referencing system being used will help eliminate these issues.

Random roughness has no pattern and its cause can be difficult to identify. The first step is to mark the rough areas in the HMA pavement and look for some type of pattern. Material being spilled on the grade and left in piles might be one cause. This HMA can segregate and cool enough to cause the screed to rise as it paves over the spilled material. Any changes in the speed of the paver should be noted. If the paver speeds up when material is available and slows down when it is not, roughness in the HMA pavement can result. Reflective roughness should always be a concern while paving over a poor base. A yielding subgrade can result in random roughness during compaction.

8.5 Joint Construction

8.5.1 Transverse or Construction Joints

Good transverse construction joints, like all good HMA joints, are straight, with a vertical face on the cold side of the joint. The cold face of the joint should be vertical. There are several ways to establish and maintain the vertical face on the cold side while traffic is rolling off the edge. The easiest and most economical method is to build a wedge of sacrificial material that ramps down to the underlining surface and stays in place until the paving operation is resumed, at which time the material in the wedge is removed. There are three methods of properly building the sacrificial wedge:

- In the most widely used method, the paver is stopped and the screed is lifted over the head of material in the auger area. The excess material, which is to be used to build the wedge, is then separated by shoveling a trench in the hot material just downstream from the joint. The side of the joint that remains in place should be pushed back with a lute and a sharp vertical edge built. A bond-breaking material (plastic, Kraft paper, or sand) should be used to line the bottom of the wedge area and cover the vertical face (see Figure 8.24). After the area is lined with a bond breaker, the excess HMA can be moved onto the



Figure 8.24 Placement of Kraft Paper Bond Breaker on a Transverse Joint

lined area, then raked and shoveled into a wedge. The whole area is then compacted.

- In a variation of the previous method, a wooden block the same thickness and width as the compacted mat can be inserted after the wedge area has been lined with the bond breaker and before the wedge is built. This block will remain in place to provide a true vertical face on the cold side of the joint and facilitate the removal of the sacrificial wedge of material when paving resumes.
- In the third method, after the paver has passed the joint location, the material delivery conveyors are shut off, stopping the flow of material into the screed area. As the screed is starved of material, the thickness of the mat drops until the mat thickness is feathered out to zero. When paving resumes, the cold mat must be cut back at a point where full mat thickness exists. This cutback process is done with a power saw equipped with either a dry-cut diamond or abrasive-blade, or by running a cold-milling machine transversely across the pavement.

The following precautions apply when constructing a transverse joint:

- Wet diamond sawing *should not* be used because a slurry is produced, leaving a dust film on the joint's vertical face and staining the pavement.
- Cold milling in the longitudinal direction to remove the wedge *should not* be allowed because the cutting mandrel cannot produce a vertical face.
- Bond breakers and wooden blockouts should not be used on airfield pavements opened to aircraft traffic.

To resume paving from a transverse joint, the wedge and any bond breaker material must be removed and properly disposed. If the bond breaker worked well, this should not be a difficult task. Once the wedge is removed, the area must be clean and dry before tack is applied to both the vertical face of the joint and the surface to be paved (see Figure 8.25). The tack on the vertical face will allow the hot side of the joint to adhere to the cold side and seal out moisture and contaminants that could cause the joint to ravel.

When constructing a transverse joint, to both stop and resume paving, it is good practice to use a 10-foot (3-meter) straightedge to check the pavement leading into and away from the joint for smoothness before compaction begins. If this check is done while the material is still hot, corrections can be made by adding or removing material to achieve acceptable mat smoothness.

If the temporary wedge is going to be used by traffic for only a short time (overnight), the application of the tack coat for the wedge can be eliminated, which will greatly facilitate removal.

8.5.2 Paving Start-up

During paving start-up procedures, the screed should be aligned with the cold joint being matched or centered on the lane to be paved, and then lowered onto wood blocks or laths stacked to the thickness of the

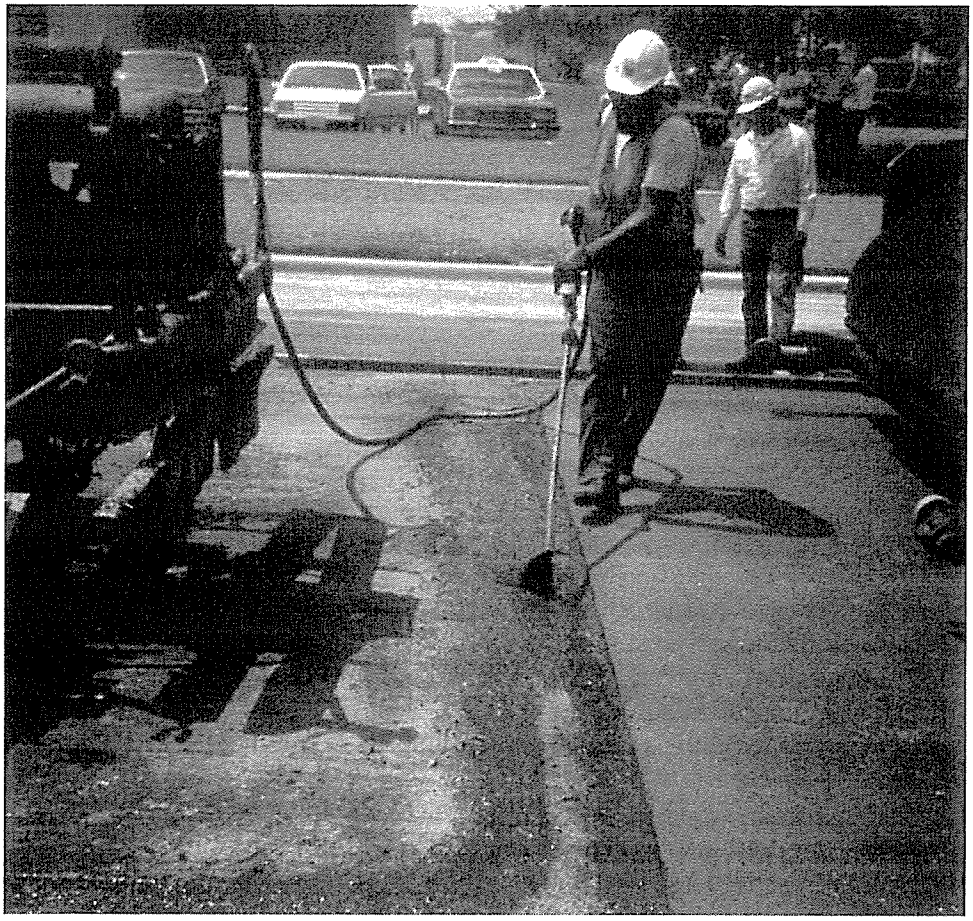


Figure 8.25 Applying Tack Coat on a Transverse Joint

new mat to be laid plus the usual 25 percent allowance for rolldown. On a previously laid mat, the starting blocks are placed on the cold mat and only need to be as thick as the anticipated rolldown. The use of these "starter blocks" holds the screed flat at its anticipated equilibrium level as it moves into fresh HMA in front of the screed. This minimizes the screed's rising or falling (introducing roughness) and significantly shortens the length of travel required for the screed to find equilibrium.

To prevent the mixture from sticking and tearing, the screed is heated to its operational temperature, around 250°F (121°C). The metal areas of the material handling system that come into contact with the HMA should be sprayed with an approved release agent before charging the hopper and the screed. This will keep the asphalt from sticking to the colder metal surfaces and help the HMA flow through the paver. The tow-points on both sides of the paver should be manually centered so they have maximum travel to provide necessary depth correction.

The screed angle of attack should then be zeroed or nulled out by turning the depth screw on both sides of the screed until a free-rotary

movement of one-half to three-quarters is achieved. Both sides should be checked to ensure that the screed is resting flat on the starting blocks. Then, the paving angle of attack should be induced by turning the depth screws in the appropriate direction. Previous experience with a similar mix will make setting the angle of attack easier. If automatic grade and slope controls are used, the grade sensor should be positioned at the correct vertical distance off the grade reference. Once this is done, the null or neutral button on the grade controller should be pushed to finish setting the automatic grade sensor. After setting the angle of attack, and with the proper slope dialed in, the slope control can be nulled out with the reset button.

The hauling unit is then directed into position and material discharged into the hopper. The auger conveyor is turned on and monitored as the material fills the auger area. If material does not automatically fill the outside corners, mix can be shoveled out to the ends to fill the screed properly. The material sensors should be checked to ensure they are operating properly and set to index off the live head of material.

After the paver pulls off the joint, the thickness should be checked repeatedly with a depth probe over the first 25-foot (7.6 meters) and a slope board used to double-check the specified slope. As soon as true grade and slope have been established, the controls should maintain the desired profile.

8.5.3 Longitudinal Joints

A poorly constructed longitudinal joint can greatly affect the life of an HMA pavement. Joint deterioration starts when air, water, and contaminants find their way into the joint, through either segregation, poor density, or the two mats not bonding properly. As the joint deteriorates, it will require joint sealing, patching or premature resurfacing. The following suggestions will greatly improve the chances of constructing a longitudinal joint that is free of segregation, compacted to specified density, and well bonded to the previously laid HMA pavement:

- Steer a straight line when laying the first pass
- Create as close to a near-vertical edge on the cold-side of the joint as possible
- Minimize segregation on both outside edges of the screed.
- Apply a tack coat or other approved sealant to the cold side of the joint
- Ensure proper overlap of the previously laid mat
- The hot side of the joint (second pass) should have sufficient depth to allow proper rolldown and a smooth match with the cold side
- Minimize the work done by the lute or rake man. Only if needed, the hot material should be bumped back to the hot side of the joint and not broadcast across the hot mat
- Compact for smoothness and density

Steering a straight line is necessary for constructing a sound joint. When the matching pass is laid, the overlap on top of the cold mat should be ½–1 inch (12.5–25 millimeters). This overlap will be difficult to achieve if the first pass is not straight. The screed operator can help maintain this close tolerance by moving the screed in or out—if the paver is equipped with a variable-width screed.

Controlling segregation on both outside edges of the screed can be done with a proper setting of the material sensor controlling the flow of material. Auger and tunnel extensions can also be used to reduce the segregation of the material being moved outward by the augers. Observation is the key. Any segregation observed along the outside edges of the screed requires corrective action (see Figure 8.26).

The proper depth needs to be laid to allow for rolldown to smoothly match the adjacent compacted mat. The typical amount of rolldown is about 25 percent. For example, a 2-inch (50-millimeters) HMA pavement



Figure 8.26 Segregation Along Outside Edge of Mat

typically requires 2½ inches (63.5 millimeters) of uncompacted HMA material for a correct rolldown. This will vary some depending on the mix stiffness, with stiffer mixes having less rolldown. After the joint has been placed properly, compaction can begin. The rolling options are discussed in Chapter 9, "Hot Mix Asphalt Compaction."

8.5.4 Notched-Wedge Joints

The notched-wedge joint was developed to help prevent premature failures of longitudinal joints. The notched wedge (see Figure 8.27) can improve the joint structure over the vertical edge joint by increasing the bond surface area between the two mats. The initial pass is laid with a 1-foot (300-millimeter) taper with a built-in ½- to 1-inch (19–25 millimeter) notch or vertical face at the end of the taper. The notch prevents the fracturing of aggregate when the joint is rolled by preventing the wedge from being tapered down to zero.

Another plus for the notched wedge is traffic safety after the first pass is laid. The notched wedge provides a ramp for traffic to get safely from one side of the joint to the other. Because of this safety feature, paving can continue the entire day on one side of the pavement and move back the next day to pull the lanes even, instead of moving back at mid-day. The screed extension that lays the notched wedge is available from several manufacturers whose operational procedures must be followed to produce a sound, durable longitudinal joint.

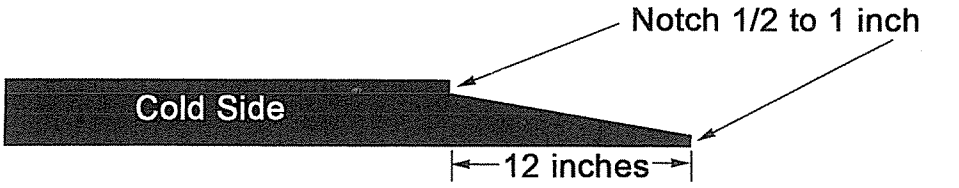


Figure 8.27 Notched-Wedge Joint

8.6 Related Paving Operations

8.6.1 Hand-Spreading Operations

When the major portion of a project will be placed manually without the aid of a paving machine, mixes should be designed for good workability. A good mix design technician has several options for making mixes that are easier to place manually.

Placing and spreading by hand should be done very carefully, with the material distributed uniformly to avoid segregation of the coarse aggregate. When the HMA is dumped in piles, it should be placed in a manner where the shovelers have to move the least amount of material the least distance. Sufficient space should be available so that workers are not required to stand in the HMA.

HMA should not be broadcast or spread from shovels because segregation will result. Rather, it should be deposited from shovels into small piles and spread with lutes. Any part of the HMA that has formed into lumps and does not break down easily when struck with a shovel edge should be discarded. After the material has been placed, and before rolling starts, the surface should be checked with straightedges, and all irregularities should be corrected. On cool days, careful attention must be paid to the HMA temperature. Rolling should begin as soon as possible to ensure adequate compaction.

8.6.2 Supplemental Operations and Appurtenances

Items such as roadway widening, shoulders, superelevated curves, curbs, ditches and revetments are considered supplemental paving operations. In some cases, such as roadway widening and shoulder construction, specially designed pieces of equipment and attachments allow the work to be done easily and efficiently. Certain roadway appurtenances, such as asphalt curbs, dikes, ditches, spillways, and slopes, are increasingly becoming part of paving contracts. For these jobs, special paving equipment or special HMA mixtures (or both) may be required.

Roadway Widening and Shoulders

Two-lane pavements today should be 24–28 feet (7.3–7.9 meters) wide. Many thousands of miles of primary routes are now inadequate in width and thickness. Narrow roads should be widened before they are strengthened with overlays. Skillful planning can greatly improve the original alignment. One technique is to place most or all widening at one side or the other around curves. For example, sharp curves should be widened

on the inner side to 30 feet (9.1 meters) and superelevated. Widening may range from a few feet (>1 meter) on one side of a pavement to the addition of full-width traffic lanes on both sides.

Traffic movement must be considered throughout construction. Work should not take place on both sides of the pavement at the same time, and excessive lengths of open trench should not be allowed.

Widening requires trenching to adequate width and depth, typically by templates or bits attached to a motor grader so that the walls and base are neat and true to line and grade. The subgrade is compacted to the required density or at least rolled until the bottom is firm and free of loose material. Existing pavement edges are trimmed to a nearly vertical state and cleaned of all dirt and foreign material. The edge is then sprayed with ample tack coat to help adhere the new HMA to the existing pavement. The HMA base is placed in layers and compacted to the required density. Special trench rollers are available in different widths for this.

When shoulders are added to an existing paved roadway, the same procedures and precautions for roadway widening are observed. It is important that the new shoulder aids, rather than retards, surface-water runoff. Also, the vertical face of the existing pavement should be given an adequate application of tack coat to seal the joint.

When paved shoulders are part of the initial construction of a roadway, procedures for paving travel lanes should be followed.

Superelevation

Paving curve sections with superelevated cross slopes on new construction does not generally pose a problem because the prepared foundation material is usually superelevated or sloped the same as the new pavement. This allows the same uniform thickness that is being placed on the tangent or straight sections to be placed on the curved or superelevated sections. As the paver proceeds through the transition to the superelevated section and out again, the paver and the screed tilt accordingly on the prepared foundation, and little if any adjustment in the screed controls is required.

If the superelevation is not built into the prepared base or subgrade, then it must be built into the HMA pavement with variable thickness layers. Today's pavers are equipped with automatic slope controls, allowing the transverse cross slope to be dialed in as the paver moves through the superelevated curve. Sensors riding on an erected or traveling stringline maintain grade control for the end of the screed nearest the sensor. The depth at the other end of the screed depends on the transverse slope setting. When the paver reaches the point where the superelevated slope begins to change, stakes or other markers are placed on adjacent lanes with the required cross-slopes settings so that the change is slowly dialed in as the paver proceeds through the transition. The change in slope is gradual, and small errors in dialing do not affect the riding quality of the pavement. Once the paver enters the full superelevated cross slope, the slope setting will remain the same throughout. Then, as the paver exits the fully superelevated section, the cross slope is gradually dialed out using the same procedure, but in reverse.

In building up the superelevation of curves with HMA, wedges are constructed in lifts that taper from thick on the outside edge to thin on the inside edge. Successive layers also increase in width. For instance, Figure 8.28 shows a 6-inch (150 millimeters) lift made up of three successive layers 8, 10, and 12 feet (2.4, 3.0, and 3.7 meters) wide for a 10-foot (3.0 meters) lane.

Asphalt Curbs

Asphalt curbs (dikes) are accessories to new road and street paving. On dual-lane turnpikes and interstate highways, asphalt curbs are often constructed at the outer edge of the paved shoulders on elevated grades to prevent slope erosion.

Asphalt curbs are constructed with automatic curbing machines (see Figure 8.29). These machines lay, compact, and finish straight or curved curbs for streets, traffic islands, raised medians, and parking areas. No forms are necessary. Various interchangeable mold patterns provide a wide

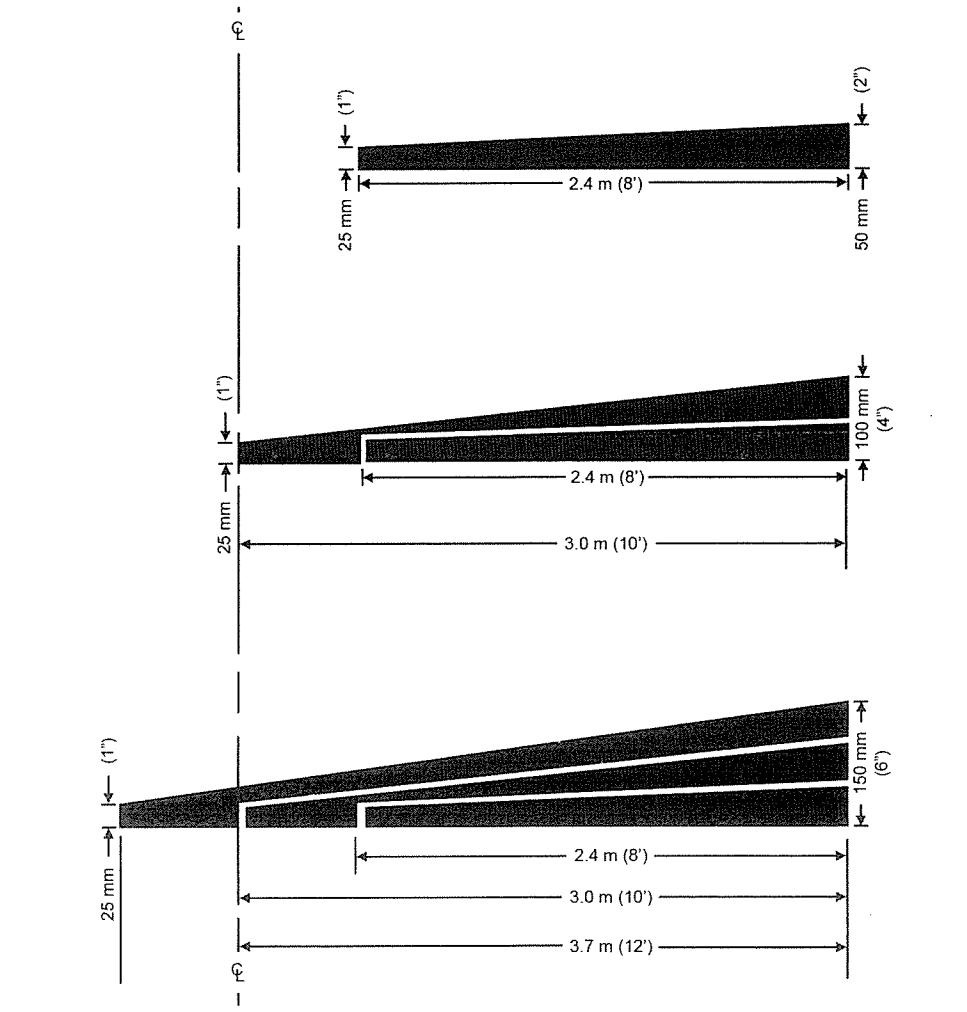


Figure 8.28 Building Superelevated Curves with HMA



Figure 8.29 Asphalt Curb Machine

selection of curbing shapes. HMA is placed in the open hopper of the curbing machine. A motorized worm gear pushes the HMA through the mold form under pressure. This pressure provides compaction and also causes the machine to move forward 10–15 feet (3.1–4.6 meters) per minute on a straight run. The production rate will depend on the stiffness of the HMA being used and the operation of the curb machine.

A fine-textured, well-graded HMA surface mixture is required for asphalt curbs. Asphalt curb HMA differs from regular paving HMA in that the quantity and the grade of asphalt cement are important factors. Curb and gutter HMA mixtures perform better when a stiffer grade of asphalt cement is used, and at a higher content (0.5–1.0 percent higher) compared to normal paving mixtures. This additional asphalt increases durability and facilitates compaction.

The usual layout practice is to place string or chalk lines on the existing pavement to guide the curb machine. A thin coating of tack is applied to the pavement surface to ensure that the curb is bonded to the pavement.

The temperature of the HMA at the time of placement is very important. If the HMA temperature is too low, adequate compaction is impossible to obtain. If the HMA temperature is too high, the curb may slough off. Normally, a temperature range of 250°–285°F (120°–140°C) is satisfactory. Machine-laid work usually does not require additional compaction. But in areas where inadequate compaction is evident, the mixture should be

adjusted, or the forward movement of the machine should be retarded by braking, or adjustments should be made to provide adequate compaction.

When used as a curb stop for vehicles, HMA curb must be backfilled before opening to traffic to prevent excessive displacement of the curbing.

Asphalt Ditches and Spillways

Ditches lie parallel to the highway at the foot of large cut sections or between dual-lane pavements having a depressed median strip. In many locations in rolling terrain, a hard impervious surface is required to prevent costly erosion. In deep-cut sections, it is often desired to construct ditches over the tops of slopes and around the ends to prevent slope erosion. Spillways to carry off water collected by curbs or dikes are subject to high water velocities and may need to be paved. In all cases, such channels should be sufficiently wide and deep so that the heaviest anticipated runoff is carried away smoothly. In some instances, very large outfall ditches may be necessary to carry runoff completely away from the right-of-way.

HMA makes an excellent ditch or spillway-lining material. As a rule, highway-surface course HMA need only be modified by increasing the binder and mineral filler contents as necessary to ensure a watertight and durable lining. The resulting lower stability is not a concern; it actually adds to the capacity of the lining to adjust to minor settlement without cracking.

Special slip-form paving machines are available for placing at least part of the ditch linings. Spreader boxes, paving machines, and even hand-placing procedures may be used. Linings are made as thick as necessary for the use and stresses involved, but practical experience indicates that a finished thickness of 1½ inches (40 millimeters) of HMA should be the minimum thickness specified. Compaction of HMA in ditches and spillways is accomplished by hand tamps, weighted slip forms, small vibrating plate compactors, or small rollers.

Roadway Shoulder Rumble Strips

Rumble strips were first used on New Jersey's Garden State Parkway in 1955. They provide a warning for drivers when their vehicle begins to run off the road. Rumble strips are continuous bands of raised material or indentations that are formed or grooved in the shoulders. They alert drivers by transmitting sound and vibration through the vehicle.

The two most widely used methods for installing rumble strips in an asphalt pavement are milled-in or rolled-in.

Rolled-in rumble strips are created by using a steel-wheel roller (see Figure 8.30) to which half-sections of metal pipe or solid-steel bars are welded. The compaction equipment presses the shape of the bar into the HMA shoulder surface. The indentation is about 1 inch (25 millimeters) deep, 2 to 2.5 inches (50 to 63 millimeters) wide and 18 to 35 inches (450 to 900 millimeters) long, transverse to the travel direction. The indentations are usually set 8 inches (200 millimeters) on center and offset from the travel lane edge by 6 to 12 inches (150 to 300 millimeters). See Figure 8.31 for a picture of rolled-in rumble strips.

Milled-in rumble strips are created by cutting (or grinding) the pavement surface with carbide teeth attached to a 24-inch (600-millimeter) diam-

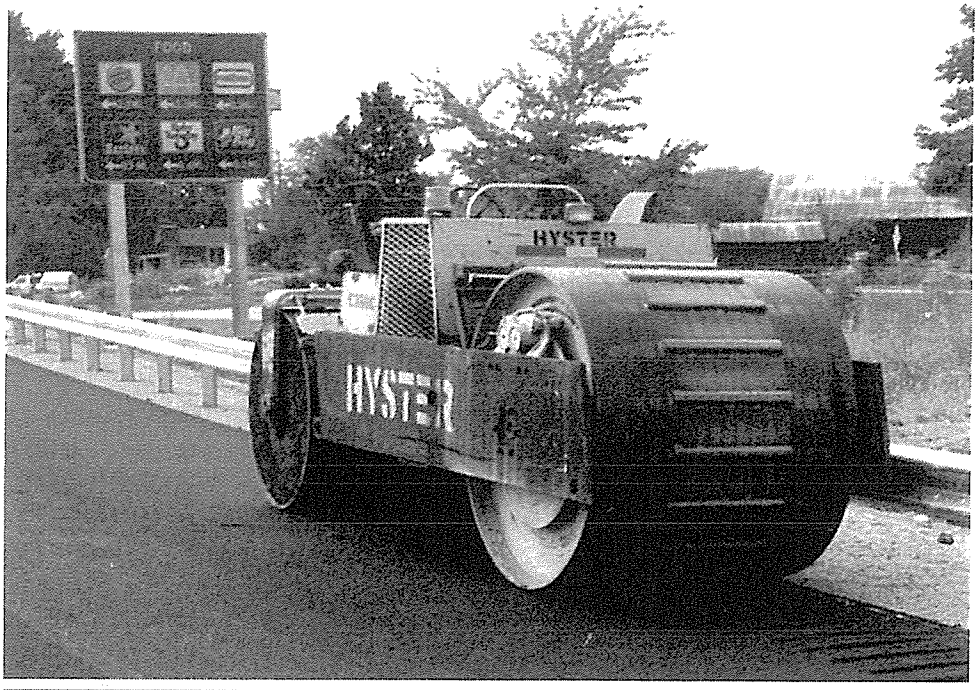


Figure 8.30 Roller Mounted with Pipe for Rolled-In Joints

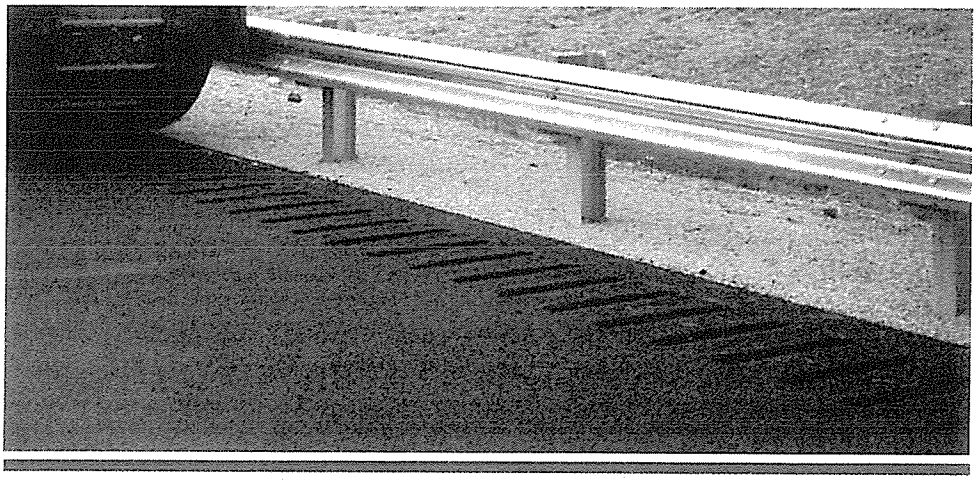


Figure 8.31 Rolled-In Rumble Strips

eter rotating drum. The indentations that are formed are approximately 0.5 inch (13 millimeters) deep, 7 inches (180 millimeters) wide and 16 inches (400 millimeters) long, perpendicular to the travel lane. The indentations are approximately 12 inches (300 millimeters) on center and offset from the edge of the travel lane by 4 inches to 12 inches (100 to 300 millimeters). See Figure 8.32 for a picture of milled-in rumble strips.

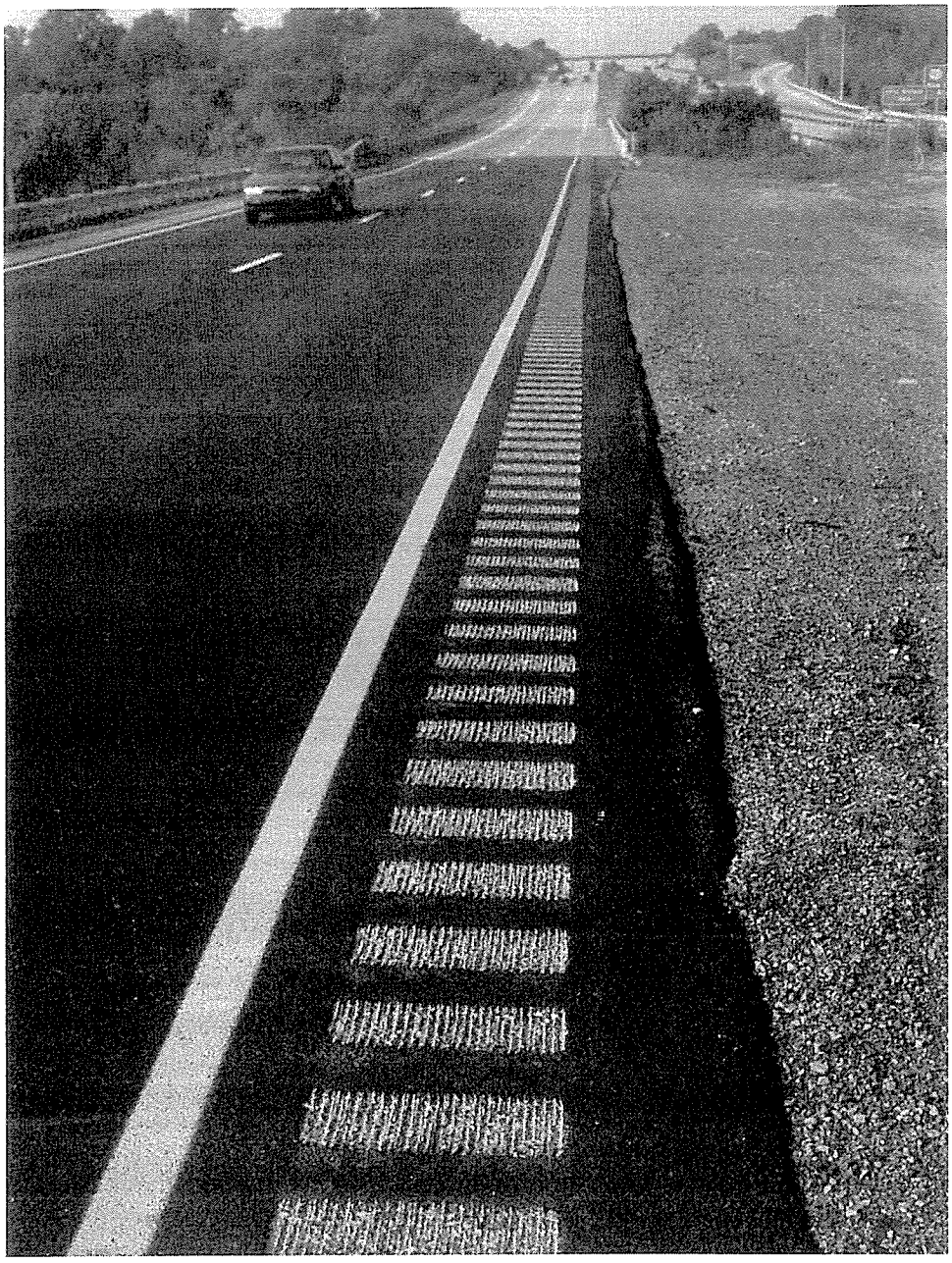


Figure 8.32 Milled-In Rumble Strips

The rolled-in method was the original method used for constructing rumble strips. Steering a straight line with a large steel-wheel roller and getting uniform indentation as the HMA cools at different rates are the two major drawbacks to this method. Multiple small milling heads mounted on rubbert-tire machines have automated the process and have become the predominate method of installation.

8.7 Foreman's/Inspector's Check Sheet

The following list includes important points to keep in mind for the project foreman, or inspector, who is responsible for placing an HMA pavement.

To build a smooth pavement:

- Never run the hopper empty between loads (The level of material in the hoppers should never be allowed to fall below the bottom of the flow gates.)
- Establish and maintain a continuous paving speed. (Establish the speed by balancing the delivery of material with the compaction process.)
- Control the head of material in the auger area to within ±1 inch.
- Do not allow trucks to bump the paver.
- Do not dump excess material on the grade (When dumping directly into the paver hopper or cleaning truck tailgates.)
- Do not fold the hopper wings into an empty conveyor system. (Fold hoppers often enough while moving forward to keep HMA at placement temperature.)
- Practice good starting and stopping techniques of the paver.
- Watch for overcorrecting of the depth screws.
- Make sure the edger plate is adjusted properly.

To make good longitudinal joints:

- Strive to minimize segregation at the outside edges of the new HMA.
- Steer a straight line on the first lane.
- Utilize the proper overlap of the cold HMA pavement being matched.
- The depth of the HMA pavement being laid should be approximately 25 percent greater in depth than the HMA pavement being matched for rolldown.
- Watch that material is not allowed to flow out or under the edger plates. (This excess material can make getting a good joint impossible.)
- If luting is required, push material back to the hot side of the joint only and do not cast material across the mat.
- Use proper rolling procedures for compaction. (See Chapter 9.)

To make good transverse joints:

- Ensure proper shutdown on original pass. (The HMA pavement left must be the proper depth and slope. Check it with a straightedge.)
- Construct a temporary wedge using some type of bond-breaker material.
- Remove the wedge, and prepare the area where the joint is to be made. (Clean all dirt, loose material, and tack the area.)
- Align the paver with the previously laid mat.

- Heat the screed properly.
- Use starting blocks. (To allow for proper rolldown.)
- Null out the screed properly.
- Fill the augers properly. (Shovel the material into the corners.)
- Begin paving and check depth and slope; correct if necessary.
- Roll the joint. (See Chapter 9.)

Resources

1. *Hot-Mix Asphalt Paving Handbook 2000.* National Asphalt Pavement Association. Lanham, MD.
2. MS-5, Introduction to Asphalt. The Asphalt Institute. Lexington, KY.
3. MS-22, *HMA Construction.* The Asphalt Institute. Lexington, KY.
4. Roberts, Freddy L., Kandhal, Prithvi S., Brown, E. Ray, Lee, Dah-Yin, and Kennedy, Thomas. *Hot Mix Asphalt Materials, Mixture Design and Construction.* NAPA Education Foundation. Lanham, MD.

CHAPTER 9

HOT MIX ASPHALT COMPACTION

This chapter discusses the mechanics of HMA compaction, the factors affecting compaction, compaction equipment, rolling procedures, and troubleshooting compaction problems.

Section 9.1 **Introduction**

Section 9.2 **Compaction Basics**
9.2.1 Terms and Definitions
9.2.2 Need for Compaction
9.2.3 Mechanics of Compaction
9.2.4 Factors Affecting Compaction
9.2.5 Cool Weather Paving

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9.3.1 Static Steel-Wheel Rollers
9.3.2 Pneumatic-Tire Rollers
9.3.3 Vibratory Rollers
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9.4.3 Lift Thickness
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9.4.5 Sequence of Roller Operations
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Section 9.5 **Troubleshooting**
9.5.1 Pickup
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9.5.4 Roller Marks

9.1 Introduction

Compaction is the process used to densify, or reduce the volume of, a mass of material. For hot mix asphalt (HMA), compaction locks aggregate particles together to provide stability and resistance to deformation while simultaneously reducing the permeability of the mixture and enhancing its durability.

This chapter will discuss the mechanics of compaction, factors affecting compaction, compaction equipment, rolling procedures, and troubleshooting compaction problems.

9.2 Compaction Basics

9.2.1 Terms and Definitions

The followings terms and their definitions are used in this chapter:

- **Compaction** is the process of compressing HMA into a smaller, denser volume.
- **Density** is a physical measurement typically expressed as a percentage of maximum theoretical density or as a percentage of laboratory density. Density is what is achieved through the compaction process.
- **Lift.** An HMA pavement consists of a series of lifts or layers of HMA that will constitute a surface, intermediate, or base course. Each course may have multiple lifts or layers. For example, a 6-inch HMA base course may have two lifts 3 inches thick or three lifts 2 inches thick.
- **Time available for compaction** is the time it takes for the mat to cool below the point where density is usually obtained.
- **Pass** is defined as the roller passing over one point in the mat one time.
- **Coverage** is defined as the roller making enough passes to cover the complete width of the mat being placed. Repeated coverages are applied until the target density is achieved. This is demonstrated in Figure 9.1, which illustrates the pass and coverage.

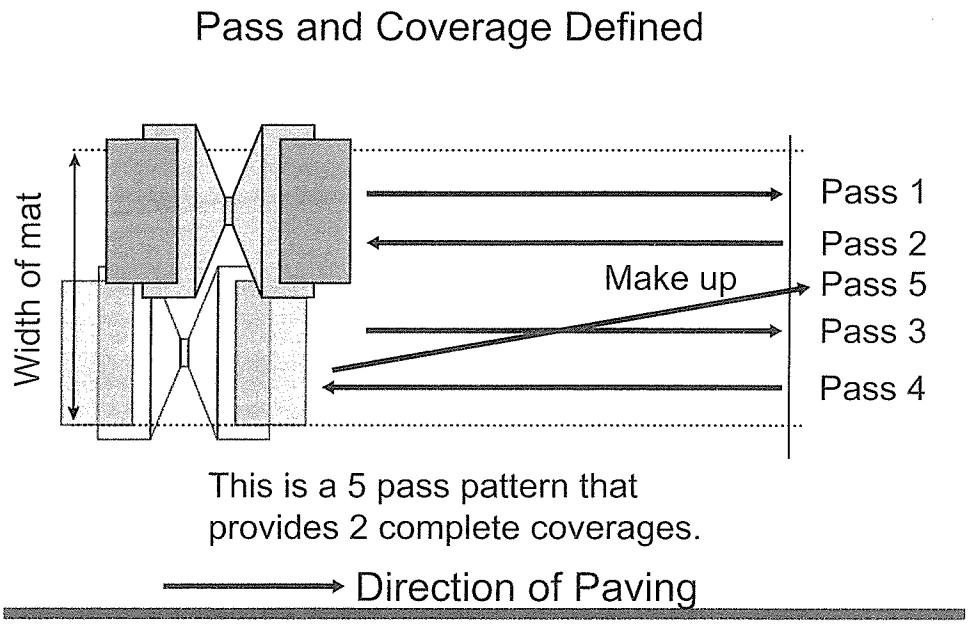


Figure 9.1 Pass-Coverage Defined